

DIGITAL FORENSIC EXAMINATION REPORT

UNMANNED AERIAL VEHICLE (UAV) FLIGHT DATA & TELEMETRY EXTRACTION

PART I: EXECUTIVE INCIDENT SUMMARY (NON-TECHNICAL & JUDICIAL BRIEF)

1. INVESTIGATIVE CASE ADMINISTRATION & PLAIN-LANGUAGE EXECUTIVE SUMMARY

Case Reference:	CASE-PX4-EXT-001	Evidence Item ID:	real_px4_flight.ulg
Examining Agency:	Cyber Forensic Investigation Laboratory (CFSL / State Police)	Lead Examiner:	Inspector Cyber
Examination Date:	2026-09-06 15:24:47 UTC	Forensic Engine:	Pushpak Indigenous Drone Forensics Toolkit (v1.0.0-phase1 (Build 2026.09))

EVIDENTIARY INTEGRITY 100% AUTHENTIC Dual SHA-256/BLAKE3 matched. Zero tampering.	AIRSPACE COMPLIANCE COMPLIANT (CLEAR) Operated in permitted boundaries.	FLIGHT ENVELOPE 2.9m 113m ALT Peak 104.4 km/h across 859 pts.	TERMINATION REASON RETURN-TO-HOME (RTH) Recorded flight disarm.
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Incident Narrative (Plain Language): On 2026-09-06, digital flight evidence for case **CASE-PX4-EXT-001** was acquired for forensic analysis. Physical and telemetry artifacts confirm the aircraft is a **PX4** UAV (Model: *PX4_FMU_V6X*). The recorded flight operated between **2026-09-05 05:42:27 UTC** and **2026-09-05 05:45:19 UTC**, remaining airborne for a duration of **2.9 minutes** across **859 recorded positions**. The aircraft attained a peak altitude of **113.0 meters (371 ft AGL)** and a maximum ground speed of **104.4 km/h**. The aircraft operated within authorized flight bounds with no restricted boundary infringements detected. Flight conclusion was recorded as a **commanded Return-to-Home (RTH) procedure**.

Key Question for Legal / Judicial Review	Plain-Language Finding & Significance
1. Was the evidence altered or tampered with?	NO. Bitstream cryptographic verification (SHA-256 & BLAKE3) confirms 100% data authenticity with zero post-seizure tampering.
2. Did the drone enter restricted airspace?	NO. The flight remained fully within authorized, unrestricted airspace corridors.
3. Who or what was piloting the drone?	The flight operated primarily under AUTO_MISSION, AUTO_LOITER, AUTO_RTL , establishing active operator command and telemetry control.
4. What caused the flight to conclude?	Flight termination was triggered by Return-to-Home (RTH) .

PART II: TECHNICAL TELEMETRY & CRYPTOGRAPHIC EVIDENCE DOSSIER (EXPERT AUDIT)

2. EVIDENCE INTEGRITY & CHAIN OF CUSTODY

Source Evidence File:	C:\Users\pawan\Desktop\dfs\sample_evidence\real_px4_flight.ulg
File Size:	27,388,166 bytes
SHA-256 Digest:	b8abd99f208de7fd072d2e271524507b761e061e6cda9170685caf85901ca027
BLAKE3 Digest:	f6f26e52465feb6e27339cf256dde2a6469252f44085e80616e1709ac3ff46dc
Chain Verification:	VERIFIED INTACT (100% Cryptographic Match)

■ WHAT THIS MEANS IN PLAIN ENGLISH

Every electronic file has an unforgeable digital fingerprint called a cryptographic hash. Because both the SHA-256 and BLAKE3 mathematical digests match the baseline recorded at the time of seizure, the court and investigating officers can be 100% certain that this evidence has not been modified, edited, or fabricated by anyone since seizure.

3. FLIGHT MISSION ENVELOPE & TELEMETRY PROFILE

Platform / Architecture:	PX4	Aircraft Model:	PX4_FMU_V6X
Hardware Serial No:	N/A	Total Telemetry Records:	1,928 events
Takeoff Time (UTC):	2026-09-05 05:42:27 UTC	Termination Time (UTC):	2026-09-05 05:45:19 UTC
Total Flight Duration:	171.6 seconds (2.9 mins)	Total GPS Waypoints:	859 points
Max Altitude (AGL):	113.0 meters (370.8 ft)	Peak Ground Velocity:	29.0 m/s (104.4 km/h)

■ WHAT THIS MEANS IN PLAIN ENGLISH

This section proves the physical flight capabilities and operating boundaries. The drone remained airborne for 2.9 minutes, reached a peak ground speed of 104.4 km/h, and climbed to 113.0 meters (371 ft AGL). These metrics legally establish the aircraft's operational range, airspeed, and chronological timeline.

4. FORENSIC ANOMALY FINDINGS & THREAT ASSESSMENT

Anomaly Type	Severity	Timestamp (UTC)	Location / Alt	Technical Finding Narrative
IMPACT OR CRASH	CRITICAL	05:45:09 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=38.3°, roll=-97.0°
IMPACT OR CRASH	CRITICAL	05:45:09 UTC	35.1913, 128.1476 Alt: 60.4m	Catastrophic attitude deviation / impact shock: pitch=1.2°, roll=-142.8°
IMPACT OR CRASH	CRITICAL	05:45:09 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=15.5°, roll=-131.1°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 59.2m	Catastrophic attitude deviation / impact shock: pitch=1.2°, roll=-142.8°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=-16.2°, roll=-154.8°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=-48.5°, roll=177.9°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 57.5m	Catastrophic attitude deviation / impact shock: pitch=-62.9°, roll=91.0°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=-64.8°, roll=119.5°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 55.5m	Catastrophic attitude deviation / impact shock: pitch=-62.9°, roll=91.0°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 53.1m	Catastrophic attitude deviation / impact shock: pitch=-62.9°, roll=91.0°
IMPACT OR CRASH	CRITICAL	05:45:14 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=46.1°, roll=-88.1°
IMPACT OR CRASH	CRITICAL	05:45:14 UTC	35.1912, 128.1476 Alt: 37.2m	Catastrophic attitude deviation / impact shock: pitch=14.0°, roll=-148.3°

■ WHAT THIS MEANS IN PLAIN ENGLISH

The automated forensic correlation engine identified 70 flight abnormalities. These findings establish whether the operator deliberately breached safety perimeters, disregarded critical low-power warnings, or experienced external interference during the flight.

5. FORENSIC CONFIDENCE & UNCERTAINTY ASSESSMENT

Overall Forensic Confidence Rating: DEGRADED (POOR SATELLITE COVERAGE) - 75.0%
In accordance with ISO/IEC 27037:2012 guidelines, this assessment explicitly defines the mathematical certainty and evidentiary boundaries of the reconstructed flight profile.

Forensic Dimension	Confidence Level	Evidentiary Basis & Limitations
Bit-Stream Integrity	ABSOLUTE (100%)	Dual SHA-256 + BLAKE3 hashes match original media bit-stream.
Spatial Trajectory	HIGH (75.0%)	Avg Satellites: 0.0 Avg HDOP: 0.24 Multipath risk low.
Temporal Synchronization	HIGH (99.0%)	Correlated with GPS atomic UTC reference clock.
Evidence Limitations	DOCUMENTED	Barometric altitude uncorrected for local QNH pressure.

■ WHAT THIS MEANS IN PLAIN ENGLISH

For digital evidence to be legally admissible in court, its degree of precision must be established. Because the drone maintained continuous tracking with an average of 0.0 satellites and an HDOP rating of 0.24, the flight coordinates in this report are mathematically reliable to within approximately 1 to 2 meters of true physical position.

6. CHRONOLOGICAL MILESTONE TIMELINE

Timestamp (UTC)	Event Type	Mode / State	Forensic Evidence Context
2026-09-05 05:42:27	GPS FIX	N/A	Telemetry fix
2026-09-05 05:42:28	MODE CHANGE	AUTO_MISSION	Telemetry fix
2026-09-05 05:42:28	ARM DISARM	N/A	Telemetry fix
2026-09-05 05:43:37	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: VTOL fixed-wing system failure detected
2026-09-05 05:43:37	RTH TRIGGER	N/A	[failsafe] Failsafe activated: entering Hold for 5 seconds
2026-09-05 05:43:37	MODE CHANGE	AUTO_LOITER	Telemetry fix
2026-09-05 05:43:42	RTH TRIGGER	N/A	[failsafe] Failsafe activated
2026-09-05 05:43:42	MODE CHANGE	AUTO_RTL	Telemetry fix
2026-09-05 05:43:42	RTH TRIGGER	N/A	[navigator] RTL: start return at 110 m (69 m above destination)
2026-09-05 05:44:29	MODE CHANGE	POSCTL	Telemetry fix
2026-09-05 05:45:09	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: Attitude failure (roll)
2026-09-05 05:45:09	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: VTOL fixed-wing system failure detected
2026-09-05 05:45:09	RTH TRIGGER	N/A	[failsafe] Failsafe activated
2026-09-05 05:45:11	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: VTOL fixed-wing system failure detected
2026-09-05 05:45:14	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: Flight termination active

■ WHAT THIS MEANS IN PLAIN ENGLISH

This timeline tracks the exact sequence of major decisions and events during the flight, such as motor startup, pilot control mode changes, automated alarms, and final touchdown, synchronized to international atomic time (UTC).

7. CERTIFICATE OF AUTHENTICITY (LEGAL ADMISSIBILITY)

I hereby certify under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (corresponding to Section 65B of the Indian Evidence Act, 1872) and ISO/IEC 27037 standards that the electronic records, telemetry chronologies, and mathematical calculations contained within this report were generated by lawful digital forensic acquisition. The computing systems and software tools used to parse this evidence operated properly throughout the examination with zero alteration to original electronic memory. The cryptographic digests (SHA-256 and BLAKE3) confirm absolute bit-stream integrity of the exhibited evidence.

Examiner Signature: _____

Date: 2026-09-06

Name: Inspector Cyber

Title: Lead Digital Forensics Examiner

Agency: Cyber Forensic Investigation Laboratory (CFSL / State Police)

Official Seal: [AFFIX LABORATORY SEAL]